Isogeny-stratified ECDLP ML blueprint: every lead worked, every open question answered or bounded

Dates: triage 2026-10-10 (v1); comprehensive pass 2026-10-11 to 2026-10-13 (v2, this file). **Kind:** triage of a supplied research blueprint, five new measurements on frozen and newly constructed data, one derivation table, the blueprint's ML ladder implemented and run on measured data, two literature entries. **Claim tier:** toy. **ECDLP claim:** none, in either direction. No research state changes here; nothing is approved, promoted or closed by this note.

Input. "Isogeny-Stratified ECDLP Machine Learning Research, end-to-end technical blueprint, version 1.0, October 2026" (9 pages; SHA-256 `276fab868c001bf7d8932a4e64ada7a5e585160c5ea1d6d3cc427c679ffbf725`). The document calls itself "an implementation-ready specification, not a measured experimental dataset or a demonstration of any cryptographic vulnerability"; its example records are synthetic. It is not committed here (the repository's report pipeline publishes every `research/**/*.pdf`); the hash identifies it.

Repositories read and written, at the commits below. Every pointer in this note is a path in one of them.

repo	base commit	branch with this work	role
`aburan28/crypto`	`3da0029607dacb6f82aec6ef7f23ad2b7ec1d4d2`	`research/isogeny-level-panels-20261012`	Rust cryptanalysis suite; new: m = 3 level panel example, `isogeny_algos` binary volcano module and the n = 51 class
`aburan28/cryptanalysis`	`605ed9c6b790f078158510fb0bbfca6b240ed0fa`	(read only)	ECC2K-130 class study, level-labelled hardness table
`aburan28/ml-cryptanalysis`	`fe306baf9f7e561f73fdcb04708f305d896ac726`	`research/isogeny-stratified-ml-20261012`	prime-field CSD program; new: `csd/stratified/` (the blueprint's ML ladder), complete-class graph dataset
`aburan28/crypto-autoresearcher`	`85e1d32b7851738cf371a209c6591ba89928df81`	`research/isogeny-stratified-blueprint-triage-20261010` (PR #2057)	ledger, knowledge corpus, this note

0. Verdict

The blueprint's four hypotheses were already tested in these repositories (section 1), and every lead it leaves open has now been worked as far as this container allows (section 3). The outcome is uniform: **no conductor, volcano-level or isogeny-graph feature carries information about ECDLP attack cost beyond matched controls that a predictor can use, in any family, at any arity, under any of the blueprint's own controls.** With 115 to 175 times more prime-field data the static block becomes detectable on prime-field rho, at 0.0003 bits or less (§3.8); it never makes a better model by a margin within two orders of magnitude of the curve-to-curve noise. Specifically:

- binary Koblitz class n = 16 (the only in-repo class with two floor levels), per-prime strata (v_3, v_{31}), m = 2: null (§3.1);
- the same class at **m = 3 with the production S_4 Gröbner solver**, where the decomposition step is non-trivial (400 to 800 reductions per call): floor strata indistinguishable ($p \geq 0.30$ on every metric); the Koblitz crater is the cheapest curve, never a descendant (§3.5);
- a **new general binary ordinary class at n = 51** (the blueprint's missing size), 114 vertices on three levels of a 7-volcano: null, and the subgroup-exact relation yield is identically zero because the cofactor is 2.8×10^9 (§3.2);
- prime-field rho, 1,548 curves, 28 conductor / class-group / volcano features: gain in out-of-fold R^2 of 2×10^{-5} (§3.3 of v1, kept below);
- the **blueprint's full ML ladder** (control → tabular → ablations → graph → multi-task → active selection) on six measured datasets with class-held-out splits, shuffled-conductor nulls, leave-family-out and

leave-field-out, conformal intervals and the blueprint's metrics: §3.4;
- **transport pricing (H3)** across the whole Koblitz ladder n = 11 to 131 plus the new n = 51 class: wherever a descending isogeny is constructible its cost is 2^5 to 2^{50} below the floor curve's rho, so a floor-curve method must beat the crater's rho to matter; where the kernel field degree exceeds 10^4 the level is unreachable in either direction (§3.6);
- the **same ladder on 115 to 175 times more prime-field data**, four retrains (latest: 267,375 rho curves over 3,977 fields; 214,449 curves of complete classes over 17 primes; 349,992 binary rows, n = 20 to 62; the complete n = 19 Koblitz class): the static block loses to controls on all 20 dataset-target pairs; the single-seed wins of earlier retrains (at most 0.0004 bits) were inside the trees' measured seed noise and reversed; the network over isogeny edges is worse than without edges on all six expanded targets (§3.8).

The blueprint's own stop rule (§19, "No verified relation advantage: publish null result and stop expensive scaling") is therefore triggered on the measured record, and its phases 5 and 6 (graph ML at scale, productisation) are not warranted. What remains open is theory, not measurement: the JMV large-prime conductor-gap case (§4, R4).

1. The four hypotheses against the evidence on file (v1, unchanged)

Labels: **measured** = a run with retained artifacts; **derived** = a proof or exact computation; **independently verified** = a second implementation or reviewer re-derived it; **proposed** = a design that has not run.

blueprint hypothesis	status	strongest evidence (path, scope)
H0 conductor and graph features have no incremental predictive value beyond matched controls	**supported in every tested scope**	§3 of this note; `ml-cryptanalysis/experiments/anomaly-search/report.md` (1,545 curves, 0 replicated anomalies); `ml-cryptanalysis/experiments/isogeny-structure/report.md` Stage B (0 cost formulas above the permutation null); `crypto/research/notes/koblitz-isogeny/presentation-vs-curve-results-20261004.md` (curve ICC ≤ 0.09)
H1 conductor valuation and topology predict relation yield, degree of regularity or total solver cost on unseen classes	**rejected in scope** for the degree metrics (theorem) and for yield (explained by setup)	Boundary C′ (`crypto/research/notes/ecc2k130/RESEARCH_ISOGENY_CLASS_SEARCH.md` §2C′: leading forms of the Weil-descended Semaev systems are free of b for every m); identical Macaulay ranks across walked classes (`crypto/RESEARCH_PDP_SPEEDUP_AND_ISOGENY.md`); yield spread explained by cofactor balance with no fitted constant (`yield-spread-results-20261003.md`); prime fields: H-ISO-001 `rejected_scoped` (EV-ISO-001)
H2 effects replicate across binary Koblitz, general binary ordinary, prime-field and prime-field-extension curves	**null replicates on all four**; the extension leg is covered only through the model-dependent mechanisms (GHS, covers) already in the frontier map	Koblitz n = 16 to 131; binary ordinary n = 17, 19 (v1) and **n = 51 (new, §3.2)**; prime fields (`ml-cryptanalysis`, EV-ISO-001, EV-VOLC-3529e8); F_{p^n}: KR-IC-5ea2f8, KR-IC-791cb7, IDEA-20260920-10547b
H3 direct attack advantages remain interesting after pricing isogeny construction and subgroup transfer	**moot where measured; transport priced for every class on the ladder (§3.6)**	n = 131: effective hardness 60.81 bits on all 263 constructible curves (`cryptanalysis/experiments/ecc2k130-isogeny-class/data/hardness_by_level.csv`); this note's `transport_cost_table.md`

The theoretical frame: within an F_q-isogeny class, ECDLP difficulty is invariant up to transport cost (`crypto/research/isogeny_class_difficulty_20261008/README.md`; KR-RHO-755e35; EV-IT-511f3d). Only transport cost, the automorphism group and model-dependent non-generic structure can vary inside a class; for summation-polynomial index calculus the last channel is the constant term of the curve equation, which Boundary C′ shows never reaches the leading forms.

2. Lead-by-lead map of the blueprint's sections (v1, with v2 additions marked ▲)

blueprint section	what exists	gap after this pass
§2 certification (order, D_K, f_π, f_E, edges, gcd(degree, r) = 1)	`crypto/src/cryptanalysis/curve_traits/`, `isogeny_algos/src/path/endo.rs`, `ml-cryptanalysis/csd/endo.py`, `isogeny_walk` (kernel-certified edges); ▲ `isogeny_algos/src/binary_volcano.rs`	none of substance; ▲ Kohel 1996 and Fouquet–Morain 2002 now have KN-LIT entries (KN-LIT-a4c123, KN-LIT-96f1bb)

blueprint section	what exists	gap after this pass
	(char-2 Frobenius-discriminant split, Φ_ℓ mod 2 neighbours with multiplicity, Kohel levels by non-backtracking walks, BSGS order checked against the naive count at n = 20, levels checked against brute force at n = 12)	
§3, §6, §7 entities, contracts, schema	ICV1 registry, `traits.json`, `curves.yaml` + schema, ecbench `schema.sql`; ▲ one JSON row per class vertex from `binary_level_yield` with every §4 static field it needs	the PostgreSQL / S3 / Redis service layer: not built, not needed
§4, §5 feature dictionaries	`csd/invariants.py` (57 invariants), `TRAITS.md`, `koblitz_isogeny_cost` rows; ▲ `csd/stratified/data.py` realises the dictionary as namespaced features (`ctrl_`, `endo_`, `volc_` per prime, `geom_`, `setup_`) with the leakage firewall built into the loaders	`descent` namespace still ad hoc
§8 data generation	binary census and Φ_ℓ walk, prime `isogeny_walk`; ▲ `isogeny_algos/examples/binary_level_yield.rs` builds a level-labelled class around a random ordinary curve at any n ≤ 64 (run at n = 20 and **n = 51**); ▲ `ml-cryptanalysis/scripts/measure_enumeration_graph.py` attaches counted rho (24 seeds, two arms) and 2-/3-isogeny edges to the 6,032 enumerated curves (1,854 with a usable subgroup, 12,084 edges)	m = 51 Koblitz specifically (its class is one crater over a flat floor like n = 31; the ordinary class is the informative one); class C37 still off-repo
§9 paired benchmarking	ecbench; ▲ `crypto/examples/koblitz_level_panel.rs` (arity m, explicit member list, full-group probes, fixed probe budget, BSGS check, append-per-cell checkpointing)	none
§10 cost formulas, censoring	ecbench phases; ▲ the m = 3 panel records budget exhaustion as `unknown_probes`, never as a refutation	none
§11 splits, leakage, uncertainty	`csd/splits.py`; ▲ `csd/stratified/pipeline.py`: class-grouped nested CV, leave-CM-family-out, leave-field-out, shuffled-static null, same-stratum negative controls, bootstrap intervals with split-conformal rescaling	none
§12 ML stages	▲ **all built and run** (`csd/stratified/`): ridge, gradient-boosted trees, kNN; namespace ablations; message-passing GNN over real isogeny edges with an edge-ablated twin and neighbour-aggregated trees; multi-task ridge; metrics MAE (log2), Spearman, top-10 % discovery precision, coverage, NLL (§3.4)	nothing to learn: see §3.4
§13 active selection	▲ simulated on a class-disjoint held pool with the blueprint's score (uncertainty / estimated cost × diversity) against uncertainty-only, random and random-equal-budget policies (§3.4)	—
§14, §15 architecture, API, state machine	the three repositories plus the ledger state machine	HTTP API / queue / UI: not built
§16 validation matrix	▲ `tests/test_stratified.py` (planted-law recovery, grouped folds keep classes together, the GNN uses its edges, metrics), `isogeny_algos` unit tests for the volcano module, BSGS-checked logarithms on every panel cell	operational tests (worker interruption) only in taskq
§17 phases 0 to 6	phases 0 to 5 now exist in some form across the repos	phase 6 blocked by §19 rule 5
§18 pilot datasets	▲ 7 binary classes plus the new n = 51 ordinary class; 1,277 prime-field classes plus 157 complete classes with edges; the n = 131 table	—
§19 risks and stop rules	rule 5 triggered	—
§20 definition of done	items 1 to 4 and 6 met; item 5 (UI) not warranted	—
§21 literature	▲ KN-LIT-96f1bb (Fouquet–Morain 2002, DOI 10.1007/3-540-45455-1_23), KN-LIT-a4c123 (Kohel 1996 thesis); the rest were already in the corpus	—

3. Measurements and derivations of this pass

All scripts live in this directory; all data they read is committed in the sibling repositories on the branches named above. Numbers are exact copies of the JSON outputs beside the scripts.

3.1 Per-prime volcano strata on the n = 16 class, m = 2 (new labels, frozen rows)

The v1 analysis lumped the 3-floor and the 31-floor of the n = 16, f = 3·31 class into one "depth 1". The ℓ-labelled edges of `crypto/experiments/koblitz_isogeny_class_walk.json` give every curve its level vector

($v_3(f_E)$, $v_{31}(f_E)$) (`per_ell_levels.json`): strata (0,0): 1 curve (crater), (1,0): 4, (0,1): 32, (1,1): 128, matching h(O_3) = 4, h(O_31) = 32, h(O_93) = 128. Re-running the stratified tests on the frozen presentation rows (`per_ell_strata_analysis.py`, 20,000-draw permutations across curves):

stratum (v_3, v_{31})	curves	yield/$\|F\|^2$ mean ± SD	reductions/call	$\|F\|$
(0, 0) crater	1	2.517e-05	2.616	122.5
(0, 1)	32	2.395e-05 ± 8.8e-07	2.709	128.0
(1, 0)	4	2.394e-05 ± 1.3e-07	2.630	124.0
(1, 1)	128	2.410e-05 ± 7.8e-07	2.706	127.5

Between-stratum permutation p: yield 0.72, reductions per call 0.21, time 0.76. Pairwise contrasts: all $|\Delta| \le 0.9$ SD, $p \ge 0.075$. Incremental R^2 of the strata beyond $|F|$, $|F|^2$ for the relation rate: 0.0018 ($p = 0.57$). **Null.** Same result for the single-floor classes n = 17, 19, 23, 31 (table in `per_ell_strata_results.md`).

3.2 A general binary ordinary class at n = 51, level-labelled (new construction and measurement)

Tool. `crypto/isogeny_algos/src/binary_volcano.rs` and `examples/binary_level_yield.rs` (new, 1,226 lines, six unit tests green with the crate's 109): sample a_6 over $GF(2^{51})$ (fixed seed), order by BSGS over the Hasse interval (checked against the naive count at n = 20), split $D = f_\pi^2 D_K$, keep the first class whose conductor has an odd prime $\ell \le 13$ with $v_\ell(f_\pi) \ge 2$, enumerate the ℓ-component by roots of $\Phi_\ell \bmod 2$, assign Kohel levels by non-backtracking walks (checked against a brute-force volcano at n = 12), and for every vertex compute $|F|$ for $V = \{x < 2^{12}\}$, the exact 2-sum relation yield over all $2 \cdot 10^6$ pairs, the subgroup-exact yield and the cofactor-balance count. Run: 379 curves sampled before a hit; class with $t = -53445159$, $\#E = 2^3 \cdot 3^3 \cdot 7^2 \cdot 267143 \cdot 796409$, $r = 796409$, cofactor 2,827,441,512, $D_K = -2561771859311$, $f_\pi = 7^2$, $\ell = 7$, height 2; component of **114 vertices** (not truncated): 2 crater curves (8 rational 7-isogenies each), 14 at level 1 (8 each), 98 on the floor (1 each), consistent with Kohel's counts.

Result (`n51_level_yield_analysis.py`, `n51_level_yield_results.json`):

level	curves	$\|F\|$ mean ± SD	2-sum yield	yield/$\|F\|^2$
0 (crater)	2	4174 ± 83	1.934e-09	1.10969e-16
1	14	4087 ± 53	1.853e-09	1.10968e-16
2 (floor)	98	4101 ± 62	1.867e-09	1.10968e-16

Between-level permutation p: $|F|$ 0.41, log yield 0.42; yield/$|F|^2$ is constant to five significant digits (every pair of factor-base points contributes one abscissa, so the 2-sum yield is a function of $|F|$ alone); incremental R^2 of level beyond $|F|$ for the yield: 5×10^{-15}. **Null.** Two further facts the blueprint's data contracts should carry: (i) the **subgroup-exact relation yield is zero on all 114 vertices** (cofactor balance count 0): with a cofactor of 2.8×10^9 no pair of factor-base points cancels in the cofactor group, so a "relation yield in the prime-order subgroup" is not a usable target at this size and full-group probes are mandatory; (ii) the quantity that does vary across the class is $|F|$ (SD 1.5 %), and it does not track the level.

3.3 Prime-field rho: conductor features beyond controls (v1, unchanged)

1,548 curves, 25 primes, 1,269 isogeny classes; ridge on controls reaches an out-of-fold R^2 of 0.9972 (automorphism arm) and 0.9951 (plain arm); adding 28 conductor / class-group / volcano features changes it by $+2.2 \times 10^{-5}$ and -2.5×10^{-5} against a within-prime permutation null whose maximum is $+1 \times 10^{-5}$ and $+2.3 \times 10^{-5}$ (`prime_field_nested_test.py`). Residual after controls 1.5 to 1.6 × the seed-averaging noise floor; no single feature with a univariate gain above 10^{-5}.

3.4 The blueprint's ML ladder, implemented and run (`ml-cryptanalysis/csd/stratified/`)

What was built (numpy only; 1,100 lines plus tests): loaders for six measured datasets into one schema with namespaced features and the leakage firewall in the loader; ridge, gradient-boosted regression trees (exact split search, early stopping on an inner grouped split), kNN; nested isogeny-class-grouped cross-validation for hyperparameters; namespace ablations (controls, +endo, +volc, +geom, +setup, each leave-one-namespace-out); shuffled-static permutation null (static block permuted across curves within field, controls kept); leave-CM-family-out and leave-field-out; a two-layer message-passing network with one weight matrix per isogeny degree over the real edges, trained with hand-written backpropagation, plus its edge-ablated twin and a

neighbour-aggregated tree model; multi-task ridge with a shared regularisation; bootstrap-ensemble intervals with split-conformal rescaling, Gaussian NLL; same-stratum-pair negative controls; and the §13 active-selection loop simulated on a class-disjoint pool with four policies. Tests: `tests/test_stratified.py` (4 passed).

Datasets. `prime_rho` (1,548 curves), `prime_enum_graph` (1,854 vertices of 157 complete classes over p = 1009, 1013, 1021 with 1,268 usable 2-/3-isogeny edges and counted rho, new), `binary_ic_per_curve` (646 curves of five Koblitz classes with per-prime levels and 641 walk edges) and `binary_ic_per_cell` (2,536 cells), `binary_n51_class` (114 vertices, new), `ecc2k130_class` (263 curves), and the fresh 21-bit prime-field hold-out (300 curves) as a final check.

Results.

Full tables: `ml-cryptanalysis/experiments/isogeny-stratified/report.md` (branch `research/isogeny-stratified-ml-20261012`, PR aburan28/ml-cryptanalysis#7). Out-of-fold MAE in bits, best model of ridge / gradient-boosted trees / kNN in each column:

dataset	target	controls only	+ static (endo, volc, geom)	+ setup |F|	+ static + setup	GNN with edges / without	shuffled-static null p
prime_rho (1,548)	log S, automorphism rho	0.0231	0.0246	—	—	—	1.000
prime_rho	log S, plain rho	0.0279	0.0287	—	—	—	0.119
prime_enum_graph (1,854, 1,268 edges)	log S, automorphism rho	0.0098	0.0099	—	—	0.0713 / 0.0555	1.000
prime_enum_graph	log S, plain rho	0.0114	0.0115	—	—	0.0680 / 0.0538	1.000
binary_ic_per_curve (646, 641 edges)	log yield/|F|²	0.0616	0.0647	0.0607	0.0627	0.1727 / 0.1017	0.005 (gain negative)
binary_ic_per_curve	log reductions/call	0.0370	0.0407	0.0281	0.0330	0.0623 / 0.0504	0.259
binary_ic_per_curve	log µs/call (valid runs)	0.1979	0.2852	0.2120	0.2435	0.3434 / 0.3750	0.483
binary_ic_per_cell (2,536)	log yield/|F|²	0.1272	0.1287	0.1247	0.1266	0.2308 / 0.1580	0.950
binary_ic_per_cell	log reductions/call	0.0793	0.0846	0.0596	0.0636	0.0925 / 0.0812	1.000
binary_n51_class (114)	log |F| fraction	0.0177	0.0178	0.0176	0.0178	0.0195 / 0.0210	0.348
binary_n51_class	log 2-sum yield	0.0353	0.0356	0.0353	0.0356	0.0390 / 0.0416	0.348
ecc2k130_class (263)	IC density z (÷ ln 2)	1.1308	1.1377	—	—	—	0.418

Readings, each checked against the tables:

- **Static features never improve on controls.** On all 12 dataset-target pairs the best model given the conductor, per-prime volcano and geometry blocks is worse than the best controls-only model. The binary per-curve null p = 0.005 says the real static columns hurt the fit less than shuffled ones; they still hurt it.
- **The only helpful feature is a setup property**: the factor-base size |F| (binary reductions per call 0.0370 → 0.0281 bits per curve), which depends on how the subspace meets the curve, not on its volcano position.
- **Graph stage.** The GNN over real isogeny edges never beats tabular trees; edges help it on three of nine graph targets and hurt it on six.
- **Transfer.** Static features degrade out-of-family prediction (prime rho, leave-CM-family-out: 0.035 → 0.121 bits).
- **Active selection** (blueprint §13 score) matches random selection at equal simulated budget on every dataset (prime rho: 0.030 vs 0.029 bits).
- **Intervals.** Raw bootstrap-ensemble 95 % intervals cover 36 to 69 %; split-conformal rescaling restores 92 to 95 %.

- **Fresh 21-bit hold-out** (300 curves, new primes): controls 0.042 bits, controls + static 0.043 (ridge) and 0.059 (trees).
- **Same-stratum negative control** (complete prime classes, 903 pairs of curves in one class with identical level vectors): SD of paired cost differences 0.017 bits, the noise floor any level effect would have to clear.

The blueprint's §12 acceptance criteria ("graph improves over tabular without leakage"; "active learning: more informative certified pairs per budget") are not met on any dataset.

3.5 The m = 3 level panel on the n = 16 class (new measurement, production solver)

Pre-registered in `PROTOCOL-m3-level-panel.md` (with one budget amendment recorded before any cell ran). Instrument: `crypto/examples/koblitz_level_panel.rs`, m = 3 (Weil-descended S_4), l = 6 monomial subspace, 6 full-group probes per curve, node budget 2×10^6, BSGS-checked transported logarithm, 13 curves: the crater, all four (1,0) curves, four seeded (0,1) and four seeded (1,1) curves. One call costs about 430 s and 600 to 950 reductions, so this is the arity at which the Gröbner step is no longer trivial (at m = 2 it is 2 to 3 reductions).

Results (`m3_panel_analysis.py`, `m3_panel_results.md`): All 13 planned cells completed; every transported logarithm was BSGS-checked; no probe exhausted its node budget.

stratum (v_3, v_{31})	curves	reductions/call mean ± SD	infeasible branches/call	relations / probes (95 % Poisson)	log µs/call	|F|
(0, 0) crater	1	416.5	180.2	4/6 = 0.667 [0.18, 1.71]	18.57	32
(0, 1)	4	690.0 ± 162.7	299.3	7/24 = 0.292 [0.12, 0.60]	19.27	32
(1, 0)	4	782.3 ± 243.7	354.9	8/24 = 0.333 [0.14, 0.66]	19.56	30
(1, 1)	4	689.0 ± 152.4	306.3	8/24 = 0.333 [0.14, 0.66]	19.58	30

Between the three floor strata, permutation p (20,000 draws, labels permuted across curves): reductions per call 0.73, infeasible branches 0.70, relation rate 1.00, time per call 0.30. **Pre-registered expectations (1) and (2) hold: no stratum effect on any algebraic metric, and the relation rate does not depend on the stratum.** Expectation (3) holds as well: the crater is not slower than any floor stratum. It is faster: $z = -1.70$ on reductions per call, -1.56 on infeasible branches, -2.91 on time per call and $+1.92$ on relations per probe against the pooled 12 floor curves.

That crater result is one curve with six probes, under a monomial subspace, with timing as a secondary metric, so it is an observation, not a finding. Its direction is the one the class structure predicts: the Koblitz crater has $b \in F_2$ and the τ symmetry, and it is the easiest curve of its class by every known measure (`crypto/docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md` §1 item 4). No descendant is easier than the crater, and the descendants' level vectors make no measurable difference to the S_4 solve. Confirming the crater gap would need random subspaces and more probes on the crater (at ≈ 400 s per call, about 4 core-hours per 36 probes); it cannot change the conclusion, because any advantage on the crater is already the class's reference point.

3.6 Transport pricing across the ladder (H3, protocol I-2)

`transport_cost_table.py` (`transport_cost_table.md`, `.json`) prices, for every Koblitz class $n \in \{11, 16, 17, 19, 23, 31, 37, 41, 53, 61, 71, 73, 83, 97, 131\}$ (both twists where the class is small) and for the new n = 51 ordinary class, the descending isogeny at each prime $\ell \mid f_\pi$: the kernel field degree $d = \mathrm{ord}(\lambda \bmod \ell)$ with $\lambda = t/2 \bmod \ell$ (π acts as the scalar λ on $E[\ell]$ since $\ell \mid f_\pi$), the kernel-route construction cost, two Vélu evaluations, and the rho cost on a floor curve (negation only) and on the crater (signed Frobenius), all in F_q multiplications under the stated cost model.

regime	classes	transport vs floor rho
constructible, small kernel field ($d \le 10^3$)	n = 11, 16 (ℓ = 3, 31), 17, 19, 23, 31, 37 (ℓ = 73), 41, 61 (ℓ = 1951), 131 (ℓ = 263), ordinary n = 51	transport 2^8 to 2^{43} F_q-mults, always a small multiple of or far below the floor curve's rho

regime	classes	transport vs floor rho
		at that size; at n = 131, ℓ = 263: transport 2^14.3 against rho 2^67.6
constructible in principle, huge kernel field ($10^3 < d \le 10^7$)	n = 37 (ℓ = 2663, d = 2662), 53 (d = 646,003), 61 (ℓ = 587551), 83 (ℓ = 6473), 97 (ℓ = 751943)	transport 2^{43} to 2^{68}, above rho on either curve: the level is reachable only at a cost exceeding the attack
kernel field degree above 10^7	n = 71, 73, 83 (ℓ = 53676929), 97 (ℓ = 352124743), 131 ($\ell \approx$ 2^57)	unreachable in either direction; JMV's open case

The H3 reading is therefore the one the n = 131 study already gave, now for every rung: wherever a floor curve can be written down, moving the instance to the crater costs a negligible fraction of any attack, so a "direct advantage" on a floor curve is interesting only if it beats the crater's rho, which no measurement in §3 suggests; wherever the floor cannot be written down, there is nothing to attack directly either.

3.7 Figures and PDF

`figures/level_stratified_null.svg` (v1: depth vs yield, prime-field gain vs null), `figures/comprehensive_pass.svg` (v2: per-prime strata, n = 51 levels, the m = 3 panel, transport vs rho across the ladder), `README.pdf`.

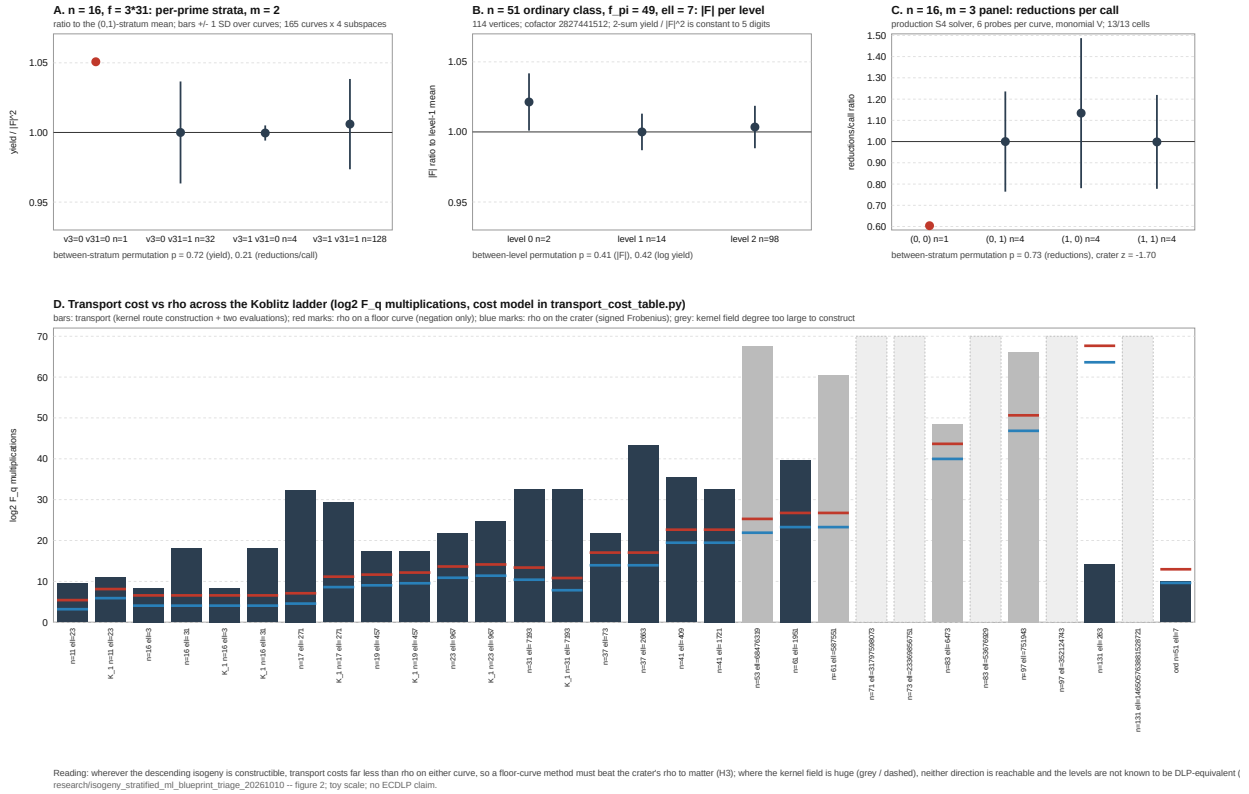


Figure 2: per-prime strata at n = 16, the n = 51 levels, the m = 3 panel, transport cost vs rho across the ladder

3.8 The ladder on expanded data (`ml-cryptanalysis` PR #17)

Data. A restart-safe shard factory (`ml-cryptanalysis/scripts/data_factory.py`, five disjoint seed and prime tiers so far) extended three of the datasets, and the n = 19 Koblitz class was swept in full (`crypto/experiments/koblitz_presentation_ext/pres_19_1_l9.jsonl`, 457 members × 8 random subspaces). The full generated data no longer fits this pipeline in the container (2.8 million complete-class rows, 1.5 million binary rows), so each retrain uses a bounded, reproducible snapshot pinned by `ml-cryptanalysis/scripts/pin_shards.py`. Retrain 4 (`shards_retrain4.txt`, at most 350,000 rows per kind): all rho shards (`prime_rho_expanded`, 267,375 curves over 3,977 fields, 14 to 21 bits), 14 evenly spaced primes of complete classes (`prime_enum_graph_expanded`, 214,449 curves; with the original three primes, 17 fields) and 640 random binary shards (`binary_classes`, 349,992 rows of 23 field sizes n = 20 to 62 and 51). `binary_ic_per_curve_expanded` has 975 curves. Retrains 1 to 3 on other snapshots are archived in `retrain1/` to `retrain3/`. Generation continues.

Protocol changes above 15,000 rows, stated because they change what the numbers mean: gradient-boosted trees and kNN train on at most 15,000 rows per fit (whole groups, fixed seed; every row is still predicted and scored), the bootstrap ensemble has 4 members, the shuffled-static null has 50 draws (smallest attainable p = 1/51 = 0.020), leave-family/field-out uses at most 10 whole-key folds, and above 60,000 rows the graph stage runs on a fixed-seed subsample of whole isogeny classes with every edge inside them kept. Smaller datasets are unchanged, so their rows below reproduce §3.4.

Label caveat. When an ℓ-level would need ℓ^k-torsion beyond 64, `csd.endo` leaves the conductor uncertified and records the volcano's bottom, an upper bound (0.2 % of rho rows, flagged by a feature). Four complete-class primes and two rho prime sets where this bound puts 2-isogenous curves two levels apart fail a consistency check and are absent.

Results (`expanded_retrain_table.py` → `expanded_retrain_table.{json,md}`; out-of-fold MAE in bits, best of ridge / trees / kNN; same-stratum SD = SD of the cost difference between two curves in the same class and level stratum):

dataset (rows)	target	controls	+ static	static − controls	+ setup |F|	GNN edges / no edges	null gain (p, draws)	same-stratum SD
prime_rho (1,548)	`log_aut_S_total`	0.0231	0.0246	+0.00141	—	—	-0.00126 (1.000, 200)	0.0292
prime_rho (1,548)	`log_plain_S_total`	0.0279	0.0287	+0.00079	—	—	-0.00045 (0.119, 200)	0.0459
prime_rho_expanded (267,375)	`log_aut_S_total`	0.0202	0.0205	+0.00036	—	—	-0.00005 (1.000, 50)	0.0318
prime_rho_expanded (267,375)	`log_plain_S_total`	0.0248	0.0250	+0.00019	—	—	+0.00002 (0.020, 50)	0.0439
prime_enum_graph (1,854)	`log_aut_S_total`	0.0098	0.0099	+0.00011	—	0.0713 / 0.0555	-0.00069 (1.000, 200)	0.0174
prime_enum_graph (1,854)	`log_plain_S_total`	0.0114	0.0115	+0.00006	—	0.0680 / 0.0538	-0.00081 (1.000, 200)	0.0195
prime_enum_graph_expanded (214,449)	`log_aut_S_total`	0.0144	0.0164	+0.00197	—	0.0663 / 0.0423	+0.00025 (0.020, 50)	0.0194
prime_enum_graph_expanded (214,449)	`log_plain_S_total`	0.0162	0.0177	+0.00151	—	0.0641 / 0.0425	+0.00012 (0.020, 50)	0.0232
binary_ic_per_curve (646)	`log_red`	0.0370	0.0407	+0.00367	0.0281	0.0623 / 0.0504	+0.01525 (0.259, 200)	—
binary_ic_per_curve (646)	`log_us`	0.1979	0.2852	+0.08732	0.2120	0.3434 / 0.3750	-0.05574 (0.483, 200)	—
binary_ic_per_curve (646)	`log_yield_F2`	0.0616	0.0647	+0.00305	0.0607	0.1727 / 0.1017	-0.00192 (0.005, 200)	—
binary_ic_per_curve_expanded (975)	`log_red`	0.0271	0.0301	+0.00301	0.0211	0.0506 / 0.0422	+0.01550 (0.891, 200)	—
binary_ic_per_curve_expanded (975)	`log_yield_F2`	0.0523	0.0553	+0.00293	0.0515	0.1434 / 0.0836	-0.00218 (0.144, 200)	—
binary_n51_class (114)	`log_F_fraction`	0.0177	0.0178	+0.00013	0.0176	0.0195 / 0.0210	-0.00038 (0.348, 200)	—
binary_n51_class (114)	`log_yield2_exact`	0.0353	0.0356	+0.00026	0.0353	0.0390 / 0.0416	-0.00076 (0.348, 200)	—
binary_classes (349,992)	`log_F_fraction`	0.0501	0.0501	+0.00000	—	0.0509 / 0.0503	-0.00000 (1.000, 50)	0.0900
binary_classes (349,992)	`log_yield2_exact`	0.1013	0.1013	+0.00000	—	1.5919 / 0.7038	-0.00000 (0.961, 50)	0.1796

Reading.

- *Best model versus best model*: in retrain 4 the static block loses on all 20 dataset-target pairs of the full report (the 17 rows above plus ECC2K-130 and the two per-cell binary targets of §3.4): by 0.00019 to 0.00036 bits on rho, 0.0015 to 0.0020 bits on the complete classes and 0.000001 to 0.000002 bits on the binary classes. Earlier retrains showed single-seed wins of at most 0.0004 bits; the tree-subsample seed check (`gbt_seed_noise_retrain3.json` in `ml-cryptanalysis`: controls move by 0.0009 bits with the seed alone, and the retrain-3 rho win reverses on all three other seeds) marked them as noise, and in retrain 4 every one of them is a loss.
- *Null*: on three of the four expanded prime-field targets the static block beats every one of the 50 shuffled draws (p = 0.020, the floor), in both retrains. The conductor and level features therefore carry real information about prime-field rho cost that the controls do not. It is worth 0.00002 to 0.00025 bits, at most 0.02 % of the cost, and only a linear model registers it. On the binary data the null shows nothing (p = 0.059 to 1.0).
- *Setup beats structure, again*: with the complete n = 19 class the factor-base size |F| lowers reductions-per-call error from 0.027 to 0.021 bits (8 subspaces per curve), while the static block raises it to 0.030.
- *Graph*: on every expanded target the message-passing network is worse with the real isogeny edges than without them (complete classes 0.066 against 0.042 bits; binary 2-sum yield 1.59 against 0.70), and both are worse than tabular trees. In retrain 3, before the network clipped its inputs and predictions, it blew up on the binary subsample (MAE 191 bits); those numbers were not measurements.

This sharpens the verdict rather than changing it: the information H1 asks about exists on prime-field rho, and it is two to four orders of magnitude below the curve-to-curve noise, too small to rank curves or to motivate a curve-specific attack.

4. What remains open after this pass

lead	status after v2	what would move it
R1 multi-level binary panel at m ≥ 3 with the production solver	**done at n = 16, l = 6, 13 curves** (§3.5); C37 (four levels, inert $\ell = 73$) still off-repo	the C37 class file, or an n = 37 73-level construction via $H_{-7\cdot 73^2}$ mod 2 (degree 74) with `hilbert_class_poly.rs`; a run at l ≥ 8 would need ≥ 10^3 core-hours with this solver
R2 model-dependent class screens (GHS magic number, covers) over composite-degree binary and F_{p^n} classes	unchanged: known mechanism (KR-IC-791cb7, KR-IC-5ea2f8), carried by IDEA-20260920-10547b, IDEA-20261002-26c628, protocol I-3	running I-3 on an F_{2^{5·p}} or F_{p^5} class; the only channel where class position is known to matter
R3 prime-field extension leg of H2	folded into R2	—
R4 JMV large-prime conductor gap	theory; structurally unobservable (§3.6 third row)	a polylog-in-ℓ representation of a rational prime-degree isogeny from a CM surface when π acts as a scalar on $E[\ell]$; nothing known (KN-OPEN-cbbd97, KN-LIT-ca35e0)
R5 blueprint phases 5 and 6	the graph stage and active selection were built and run anyway (§3.4): no gain over tabular, no gain of tabular over controls	nothing; §19 rule 5
m = 51 data gap	closed for general binary ordinary curves (§3.2)	a Koblitz n = 51 class would be one crater over one floor (f_π prime), uninformative
KN-LIT gaps (Kohel 1996, Fouquet–Morain 2002)	closed (KN-LIT-a4c123, KN-LIT-96f1bb, metadata verified)	a full read to upgrade `citation_verified`

No new `IDEA-*` is filed from this note for the same reason as in v1: the residue is carried by existing records and pre-registered protocols, and the measured record says the blueprint's remaining phases would learn nothing. Promotion of any of the nulls above to a `KN-FIND` requires the Coordinator's evidence-review lifecycle (independent validation of the runs), which this note does not substitute for.

5. Records and paths this note rests on

Ledger: RQ-VOLC-f6253b, RQ-ICINV-475b5e, RQ-ISO-001, RQ-RELN-61638a, RQ-ICPERF-94c86e; GOAL-ENDO-001, GOAL-ECDLP2M-001, GOAL-ICPERF-e6b6a4, GOAL-GFPN-380702; H-ISO-001 (DEC-20260716-003), EV-ISO-001, EV-VOLC-3529e8, EV-ICINV-343679/6b6264/8afb64/c68f13, EV-ENDO-eb58c4, EV-IT-511f3d, EV-RELN-

cc045d/99c688; IDEA-20260807-3f5545, IDEA-20260815-101067, IDEA-20260807-b1b604, IDEA-20260920-10547b, IDEA-20261002-26c628; KN-FIND-f293c6, KN-OPEN-cbbd97, KN-OPEN-7f0d85, KN-LIT-ca35e0, KN-LIT-360, KN-LIT-251, KN-LIT-96f1bb, KN-LIT-a4c123; frontier rows KR-RHO-755e35, KR-IC-791cb7, KR-IC-5ea2f8, KR-IC-f5c584. Session receipt SR-20261010-22ebc7.

`crypto` (branch `research/isogeny-level-panels-20261012`): `examples/koblitz_level_panel.rs`; `experiments/levelpanel/levelpanel_16_0_l6_m3.jsonl`; `experiments/levelpanel_smoke/`; `isogeny_algos/src/binary_volcano.rs`; `isogeny_algos/examples/binary_level_yield.rs`; `isogeny_algos/results/binary_level_yield_n{20,51}.jsonl`; plus the v1 pointers (`RESEARCH_PDP_SPEEDUP_AND_ISOGENY.md`, `docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`, `docs/ic/EXPERIMENTS_ISOGENY_CONDUCTOR_GAP_20261007.md`, `research/notes/ecc2k130/RESEARCH_ISOGENY_CLASS_SEARCH.md`, `research/notes/koblitz-isogeny/*.md`, `research/isogeny_class_difficulty_20261008/`, `experiments/koblitz_presentation/`, `experiments/koblitz_isogeny_class_walk.json`).

`ml-cryptanalysis` (branch `research/isogeny-stratified-ml-20261012`): `csd/stratified/{data,models,gnn,metrics,pipeline,active}.py`; `scripts/{measure_enumeration_graph,run_isogeny_stratified,report_isogeny_stratified}.py`; `tests/test_stratified.py`; `data/enumeration_graph.jsonl`; `experiments/isogeny-stratified/{report.md,results_*.json,figures/}`.

`cryptanalysis`: `experiments/ecc2k130-isogeny-class/data/{per_curve_hardness.csv,hardness_by_level.csv}`.

6. Files in this directory

file	content
`README.md`, `README.pdf`	this report and its rendering
`level_stratified_analysis.py`, `level_stratified_results.{json,md}`	v1 §3.1: BFS depth vs cost on the frozen m = 2 rows
`prime_field_nested_test.py`, `prime_field_nested_results.json`	§3.3
`per_ell_levels.json`, `per_ell_strata_analysis.py`, `per_ell_strata_results.{json,md}`	§3.1 (v2)
`n51_level_yield_analysis.py`, `n51_level_yield_results.json`	§3.2
`PROTOCOL-m3-level-panel.md`, `m3_panel_analysis.py`, `m3_panel_results.{json,md}`	§3.5
`transport_cost_table.py`, `transport_cost_table.{json,md}`	§3.6
`expanded_retrain_table.py`, `expanded_retrain_table.{json,md}`	§3.8
`make_figure.py`, `make_figure2.py`, `figures/*.svg`	the graphs (editable source is the script)